Global Wealth University — SmartBio Problems, Features & Implementation

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This document provides the exhaustive specification of the **Academic Problem Statement**, **Literature Gaps**, **Innovative Solved Features**, and **Technical Implementation Details** for the **SmartBio Biometric Attendance & NUC Compliance Management System**.

🎯 Part 1: Academic Problem Statement & Literature Gaps

1. The Core Academic Problem

In higher education institutions governed by the **National Universities Commission (NUC)**, students must satisfy a statutory minimum attendance threshold of **75%** in conducted course lectures to qualify for semester examinations.

Traditional attendance management suffers from systemic vulnerabilities:

- **Paper-Based Attendance Rosters**: Prone to sheet theft, illegibility, physical deterioration, and loss of records during departmental moderation.
- **Proxy Attendance & Impersonation**: Absent students frequently have peers forge signatures or call out names during manual roll calls.
- **Administrative Burden & Human Error**: Manually calculating

percentages for hundreds of students across 10–14 lectures per semester at the end of term causes delayed release of examination clearance lists.

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2. The Biometric Literature Gap (The Unreadable / Sweaty Scan Dilemma)

Existing academic literature and commercial biometric solutions introduce a critical operational failure point:

- **The Lockout Failure**: When a student arrives at a lecture hall with sweaty fingers, physical skin abrasions, dry skin, or temporary ridge distortion, the optical sensor yields a False Non-Match Rate (FNMR) or fails to match the enrolled template.
- **The Lecture Interruption Dilemma**: In conventional systems, the student is either barred from class or the lecturer must pause the lecture to perform administrative data entry.
- **SmartBio Innovation (The Flagged Exception Workflow)**:
    - SmartBio routes low-confidence and distorted scans to a real-time **Flagged Exception Queue** on the lecturer's radar.
    - The student enters the lecture hall and takes their seat.
    - The lecturer verifies the student's physical University ID card and approves the record with a mandatory **NDPA 2023 compliant audit remark**.

🚀 Part 2: System Features & Functional Modules

1. Role-Based Access Control (RBAC) Architecture with Institutional Gatekeeping

SmartBio implements a **dual-layer RBAC architecture** combining Firebase Authentication JWT tokens with Institutional Security Passcode gatekeeping at the registration level.

1a. Institutional Security Passcodes (Privilege Escalation Prevention)

Registering for any role above `STUDENT` requires a confidential institutional authorization key:

Role	Access Scope	Institutional Passcode Required
🎓 **Student**	Student Portal only	None (Open Enrollment)
👥 **Class Representative**	Class Rep · Student · Kiosk	`GWU-PROCTOR-2026`
🧑‍🏫 **Lecturer**	Lecturer Portal · Kiosk	`GWU-FACULTY-2026`
🧑‍💼 **Administrator**	All Portals (Full Oversight)	`GWU-ADMIN-2026`

1b. Navbar Route Guards (Least Privilege UI Enforcement)

Upon authentication, the top navigation bar dynamically locks down to show only portals permitted by the user's role. Restricted tabs are hidden — not just disabled — preventing any URL-based or click-based access to unauthorized views.

1c. Switch User / Sign Out

A dedicated `🚪 Switch User` button clears the Firebase Auth JWT session and returns the user to the Sign-In screen, enabling safe role-switching during multi-user demonstrations (e.g., project defense).

1d. Role Descriptions

- **Administrator**: Full academic session configuration, student and

lecturer directory, NDPA audit trail viewer, 1-click database reset, **1-Click Firestore Cloud Seeder**, and **SQL Dump Exporter**.
- **Lecturer**: Lecture session creation with real-time countdown timer, live attendance streaming radar, flagged exception resolution queue, and exportable 75% NUC defaulter rosters.
- **Class Representative (Course Proctor)**: Dedicated proctor role allowing the rep to launch the classroom kiosk terminal, view real-time hall headcount (e.g. 42/50 in hall), and broadcast defaulter warning alerts, with a strict prohibition against flag overrides.
- **Student**: Personal attendance percentage progress bars, detailed lecture breakdown logs, and **Official Printable Examination Clearance Dockets** with anti-tamper QR tokens.
- **Classroom Kiosk Scanner**: Optical sensor graphical simulation with laser sweep, minutiae detection, acoustic feedback (Web Audio API), and native browser WebAuthn biometrics.

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2. Firebase Authentication & Cloud Firestore Integration

2a. Firebase Authentication (Email/Password)

- All users authenticate via `firebase/auth` using institutional email addresses.
- JWT tokens returned on sign-in carry the user's role claims, used to enforce backend Firestore Security Rules and frontend Route Guards.
- **1-Click Cloud Seeder**: Automatically provisions all demo user accounts (students, lecturers, class rep, admin) in Firebase Authentication with `password123` during the Admin Portal seeding operation.
- **Self-Registration**: New users can register via the Sign Up screen. Accounts are created in Firebase Auth and simultaneously written to Cloud Firestore `/users` collection.
- **Password Recovery**: Institutional email-based password reset flow with role verification.

2b. Firebase Cloud Firestore (Real-Time Sync)

- **`/attendance_records`**: Live biometric scan stream — every check-in event broadcasted via `onSnapshot` WebSocket to all connected devices in < 200ms.
- **`/flagged_exceptions`**: Real-time flag queue for low-confidence scans awaiting lecturer resolution.
- **`/lecture_sessions`**: Active session state — kiosk terminals auto-detect live sessions and activate scanner mode.
- **`/users`**: Institutional directory of students, lecturers, and admins.
- **`/courses`** & **`/departments`**: Academic structure seeded from the local data repository.
- **`/audit_logs`**: Immutable NDPA 2023 compliant audit trail (write-once, no delete).

2c. Firestore Security Rules

Production rules enforce RBAC at the database layer. Test mode (`allow read, write: if true`) is used during development. Production rules restrict:

- Flag overrides to `LECTURER` and `ADMIN` roles only.
- Audit logs to write-once (no update/delete).
- User profile edits to the account owner or `ADMIN`.

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3. Dual-Mode Biometric Verification Engine

- **Modern WebAuthn API**: Interoperates with native operating system biometric authenticators (Apple TouchID, Windows Hello, Android Biometric Prompt) via browser cryptographic handshakes.
- **Interactive Optical Scanner Terminal**: High-fidelity optical sensor simulation supporting stress testing and edge-case simulations (Sweaty Finger, Injured Ridge, Non-Enrolled Stranger).

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4. NUC 75% Rule Compliance & Clearance Algorithm

- Real-time statutory attendance calculation: $$\text{Attendance Rate } (P) = \left(\frac{\text{Attended Sessions}}{\text{Conducted Sessions}}\right) \times 100$$
- Dynamic Classification:
  - 🟢 **Eligible ($\ge 75.00%$)**: Fully cleared for semester examination.
  - 🟡 **At Risk ($70.00% - 74.99%$)**: Automated deficit forecast indicating exact number of upcoming lectures required to clear the deficit: $$x = \left\lceil \frac{0.75 \times C - A}{0.25} \right\rceil$$
  - 🔴 **Ineligible ($< 70.00%$)**: Flagged as academic defaulter, barred from course examination.

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5. Nigeria Data Protection Act (NDPA) 2023 Compliance

- **Zero Raw Image Storage**: Only irreversible one-way SHA-256 minutiae hashes and WebAuthn public keys are persisted.
- **Immutable Audit Trail**: Every session creation, scan attempt, and lecturer override logs actor identity, timestamp, IP address, and security reason notes.

💻 Part 3: Technical Implementation Details

Layer	Technologies Used	Key Purpose
Frontend Core	HTML5, CSS3 Glassmorphism, Modular ES6 JavaScript	Single-Page Application (SPA), zero build toolchain, 100% GitHub Pages compatible
Authentication	Firebase Authentication (Email/Password + WebAuthn FIDO2)	JWT-based identity, RBAC role claims, session management

Real-Time Cloud Sync	Firebase Cloud Firestore (SDK v10 CDN)	Real-time WebSocket multi-device sync (`onSnapshot`) between kiosks, phones, and laptops
RBAC Gatekeeping	Institutional Passcodes + Navbar Route Guards	Prevents privilege escalation; locks UI tabs per authenticated role
Offline Storage Engine	IndexedDB & LocalStorage Repository	Offline-first relational cache with 1-click seed preloading and SQL dump generation
Acoustic Feedback	Web Audio API Oscillator & Gain Synthesizer	Tactile audio cues (laser chirp, success chime, warning beep, error buzz) with 0 external sound files
Relational Database DDL	MySQL 8.0 & PostgreSQL 16 (3NF Normalized)	Formal schema and seed scripts for Chapter 4 academic dissertation
Diagramming System	Mermaid (`.mmd` & `.md`)	Standardized flowcharts and ERDs exportable via `mermaid.live`
Mobile-First UI	CSS3 Responsive Grid, Bottom-Sheet Modals, `.form-row-2col`	All screens functional at 360px; touch targets ≥ 44px; iOS zoom prevention
Deployment	GitHub Pages (static) + Firebase Hosting	Zero-cost dual deployment; live URL for defense demo

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🔑 Part 4: Demo Credentials (For Project Defense)

Role	Full Name	Email	Password	In P
👨‍💼 Admin	Dr. Kola Balogun	`admin@smartbio.edu.ng`	`password123`	`G A`
👨‍🏫 Lecturer	Dr. Olawale Adeyemi	`o.adeyemi@smartbio.edu.ng`	`password123`	`G F`
👩‍🏫 Lecturer	Prof. Ngozi Okoro	`n.okoro@smartbio.edu.ng`	`password123`	`G F`
👥 Class Rep	Chukwudi Eze	`c.eze@student.gwu.edu`	`password123`	`G P`
🎓 Student	Benedict Uche	`b.uche@student.gwu.edu`	`password123`	*(r*
🎓 Student	Folake Adebayo	`f.adebayo@student.gwu.edu`	`password123`	*(r*